

MASENO 2025 KCSE PREDICTION



CONFIDENTIAL EXAM

451/2



COMPUTER STUDIES

PAPER 2

TIME: 2½ HOURS

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

(Kenya Certificate of Secondary Education)

INSTRUCTIONS TO CANDIDATES:

- (a) Indicate your name and index number at the top right-hand corner of each printout.
- (b) Write your name and index number on the CD/Removable storage medium provided.
- (c) Write the name and version of the software used for each question attempted in the answer sheet provided.
- (d) Answer all the questions.
- (e) All questions carry equal marks.
- (f) Passwords should not be used while saving in the CD/Removable storage medium.
- (g) All answers must be saved in your CD/Removable storage medium.
- (h) Make a printout of the answers on the answer sheet.
- (i) Arrange your printouts and tie/staple them together.
- (j) Hand in all the printouts and the CD/Removable storage medium used.
- (m) Candidates should answer the questions in English

Question One

The data in the tables below were extracted from Bondo Sub county Games Management system.

Table 1: TEAMS TABLE

Team Code	Team Name	Address	Tel. No.	Reg. Fee
01	Majiwa High School	Box 0012	021542148	5,000
02	Maranda High School	Box 2454	025485267	3,500
03	Usenge High School	Box 458	025478756	4,700
04	Akoko Mixed Secondary	Box 635	032547855	2,400
05	Kipasi Secondary School	Box 2446	032458754	1,200
06	Kapiyo Secondary School	Box 6589	015487564	1,400
07	Nyang'oma Boys	Box 1254	031204543	5,000

Table 2: RESULTS TABLE

Team Code	Games Won	Games Lost	Games Draw
01	12	2	2
02	10	3	3
03	4	8	3
04	9	3	4
05	7	3	5
06	7	6	4
07	5	9	2

(a) Create a database named **Football** to store the data above.

(12marks)

(b) Format the *Reg. Fee* field as follows:

(i) To display the entries with the prefix: “KSh.” Correct to two decimal places.

(2marks)

(ii) Restrict entries to positive values only and should return an error message “*Error: enter positive values only*” if an out of range value is entered.

(2marks)

(c) Validate Team Code to 2 characters only.

(2marks)

(d) Create a suitable table relationship between the tables.

(3marks)

(e) Design a form for each table and use it to enter the data into the tables. (9marks)

(f) Create a query named **TotalGamesQuery** to display the fields: Team Code, Team Name and Total games played.

(4 marks)

(g) Create a query to show the team name and total points. (Note: A Win in a game earns team 3 points, a draw 1 point and a loss 0). The query should show the 3 best teams based on the total points. Save the query as **BestTeamsQuery**. (5 marks)

(h) Create a report named **RegReport** to display the Team Name, Address, Reg. Fee. The report should show the total registration fee collected from all teams. (5marks)

(i) Sort the records to show the team that has paid the highest amount of registration fee first.

(2 marks)

(ii) Title the report as *“income from registration”*.

(2 marks)

(i) Print each of the following: (2marks)

(i) Teams table and Results table

(ii) TotalGamesQuery and BestTeamsQuery RegReport.

Question Two

(a) Type the following passage using a word processing package and save as **Networking**. Answer the questions that follow. (25marks)

COMPUTER NETWORK

A Computer network is a group of computers and other devices connected together by a suitable media.

Advantages of computer Networks

1. Multiple users can communicate, exchange and share information

2. Resource sharing including program, data base, hardware etc. is possible

B



By definition communications refers to the transfer of information from one place to another between two individuals using agreed symbols, signs or even human behavior such as nodding. Exchange of information relies on a communication system to convert, amplify and send signal through a common medium. Message, Sender, Medium and Receiver are the essential components of communication systems. If a system is extended by cascading more communication systems, it is called networked systems.

In case communication involves the sending of information over a significant distance, it must use telecommunications as an aid. By definition, telecommunications refers to the transmission of information between distant locations by some electromagnetic means.

Data communications is defined as the interchange and processing of encoded (digital that is, 1s and 0s) information between distant locations using telecommunications.

Data communications is regarded as the collection and distribution of the electronic representation of information which can be text, voice, graphics or image, from and to remote computing facilities. As information can only be carried to the remote site provided that the information carrier supports that particular type of data transmission, information may undergo data conversion processes if the nature of data signal is incompatible with the characteristics of the signal carrier.

Communication networks are usually defined by their size and complexity. We can distinguish four main types:

- **Local area networks (LAN).** These networks connect computer equipment and other terminals distributed in a localized area, e.g. a university campus, factory, office. The connection is usually a cable or fibre, and the extent of the cable defines the LAN.
- **Metropolitan area networks (MAN).** These networks are used to interconnect LANs that are spread around, say, a town or city. This kind of network is a high speed network using optical fibre connections.
- **Wide area networks (WAN).** These networks connect computers and other terminals over large distances. They often require multiple communication connections, including microwave radio links and satellite.

COMPARISON AMONG DIFFERENT TOPOLOGIES

Below are the advantages and disadvantages of various LAN topologies.

Type	Advantage	Disadvantage
Star	Simple and easy to identify fault	Failure of central node causes disaster
Irregular (Mesh)	Immunity to bottleneck and failure problems	Expensive to provide an alternative routing
Bus	Simple to control traffic flow	Only a single communications channel is required to service all the nodes
Ring	Simple to implement	One channel is required to service all

TRANSMISSION MEDIA

Transmission can be by cable or wireless methods. There are 3 common types of cable:

- **Twisted pair,**
- **Co-axial**
- **Fiber optic.**

Wireless transmission includes:

- Microwave transmission.
- Satellite transmission.
- Radio waves transmission.
- Infrared transmission.

TWISTED PAIR

Consists of two insulated copper wires arranged in a regular spiral pattern to minimize the electromagnetic interference between adjacent pairs

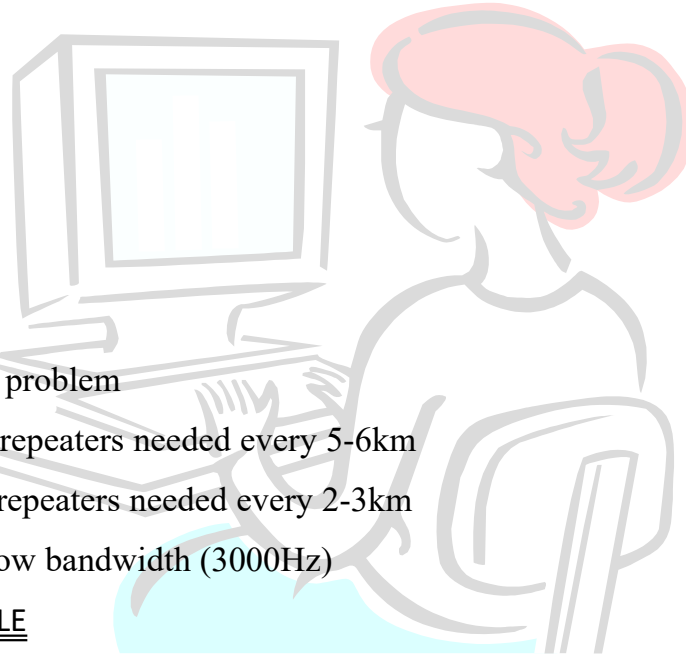
Often used at customer facilities and also over distances to carry voice as well as data communications.

TWISTED PAIR ADVANTAGES

- Inexpensive and readily available
- Flexible and lightweight
- Easy to work with and install

TWISTED PAIR DISADVANTAGES

- Susceptibility to interference and noise



- Attenuation problem
- For analog, repeaters needed every 5-6km
- For digital, repeaters needed every 2-3km
- Relatively low bandwidth (3000Hz)

CO-AXIAL CABLE

- Used for cable television, LANs, telephony
- Has an inner conductor surrounded by a braided mesh
- Both conductors share a common center axial, hence the term “co-axial”

CO-AXIAL ADVANTAGES

Higher bandwidth

- 400 to 600Mhz
- Up to 10,800 voice conversations
- Can be tapped easily (pros and cons)
- Much less susceptible to interference than twisted pair

CO-AXIAL DISADVANTAGES

- High attenuation rate makes it expensive over long distance
- Bulky
- Greater capacity (bandwidth of up to 2 Gbps)
- Smaller size and lighter weight
- Lower attenuation
- Immunity to environmental interference

- Highly secure due to tap difficulty and lack of signal radiation

FIBER OPTIC CABLE

- Relatively new transmission medium used by telephone companies in place of long-distance trunk lines
- Also used by private companies in implementing local data communications networks
- Require a light source with injection laser diode (ILD) or light-emitting diodes (LED)

ADVANTAGES OF FIBER OPTIC

- Greater capacity (bandwidth of up to 2 Gbps)
- Smaller size and lighter weight
- Lower attenuation
- Immunity to environmental interference
- Highly secure due to tap difficulty and lack of signal radiation

DISADVANTAGES OF FIBER OPTIC

- Expensive over short distance
- Requires highly skilled installers
- Adding additional nodes is difficult

(B) Required:

(i) Open a new document then copy and paste the content of Networking on the new document **(2marks)**

(ii) Save the new document as **Networking Edited.**

(2marks)

(iii) Change the title Computer Network as follows.

(4marks)

Alignment – Centre

Font Size – 23

Font colour - Red

Font type –Elephant

Kerning -12 point

Spacing –Expanded

Scale- 100%

Position- Raised

- (iv) Convert the paragraphs immediately under **Communications** into a hanging indent.(2marks)
- (v) Insert your name into the passage as a footer and Index number at the Header. (4 marks)
- (vi) Convert the paragraph under Classification **of communication networks** to two columns and justify them. (4 marks)
- (vii) Insert Water mark a cross the page the page “ **COMPUTER NETWORKING** “ (2marks)
- (viii) Change the Drop cap lines to two.
(1mark)
- (ix) Insert page border using the following setting.
(2marks)
Border – Shadow
Border – Three line
Colour - Orange
- (x) Print the document networking and Networking Edited.
(2marks)

THIS IS THE LAST PRINTED PAGE